CSC 4332 Software Quality & Testing

Abdus Satter
Spring 2026

$whoami

- PhD CS student at LSU
- MS in CS, University of California, Santa Barbara, USA
- MS in Software Engineering, University of Dhaka, Bangladesh
- BS in Software Engineering, University of Dhaka, Bangladesh
- Current research area: LLM for Security, LLM for Software Engineering

Logistics

https://abdussatter401.github.io/teaching/csc-4332.html

Book:

https://www.effective-software-testing.com

Security [Errors, Bugs, Failures]

- An error is made by a human
- As a consequence, a bug is introduced in a program
- When the bug is triggered it generates a failure
- As a consequence, the system does not function as expected

What is Testing?

“Software testing is the process of executing a software system to determine whether it matches its specification and executes in its intended environment.”

- *Testing is the process of executing a program with the intent of finding errors.* - Myers [2]
- *A successful test is one that uncovers an as-yet-undiscovered error.* - Myers [2]
- *Testing can show the presence of bugs but never their absence.* - W. Dijkstra [125]

Software Testing: Myths and Facts

Myth-1: Testing is a single phase in SDLC .

Myth-2: Testing is easy.

Myth-3: Software development is worth more than testing.

Myth-4: Complete testing is possible.

Myth-5: Testing starts after program development.

Myth-6: The purpose of testing is to check the functionality of the software

Myth-7: Anyone can be a tester.

Testing principles

- Exhaustive testing is impossible
- Knowing when to stop
- Variability is important (the pesticide paradox)
- Bugs happen in some places more than others (defect clusters)
- Testing will never be perfect
- Context is king
- Verification is not validation

Testing pyramid

- Unit Testing
- Integration Testing
- System Testing

Tool You Need for the Next Class

- **IntelliJ IDEA**